

Low Noise Power Supply With Four ICs In Parallel

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This article presents a low-noise power supply with four 78L15 voltage regulators in parallel, giving about 15V DC output. The modern environment has become noisier, and every power supply is a powerful source of electrical noise, which is highly undesirable for high-quality audio and test equipment.

If you need a power supply of +15V with a tolerance of ± 10 per cent and noise lower than 10 μ Vrms, then 78L15 voltage regulators can be used. It is because they are low-cost, easily available, have output current of 100mA or more, low output voltage tolerance, small MO package (TO-92), internal current limiting, and thermal protection.

Most 78L15 ICs have an output current of 40mA, but we may need a current of 160-250mA. The use of M series, such as 78M15 regulators with 0.5A, is not suitable because this would require complex filter circuits to minimise noisy output voltages. Also, these regulators are available in TO-220 packages, which are more difficult to mount as compared to TO-92 packages.

One of the solutions to get higher current output is to use 78L15 voltage regulator ICs in parallel. An advantage of this configuration is that if one of the parallel regulators fails, the remaining regulators will provide output voltage and current. But note that 78LXX regulators have large output noise, sometimes more than 100 μ Vrms in the range of 10Hz to 100kHz.

Generally, the noise produced by the voltage regulators is random. So, the averaging method—a noise reduction technique—can be applied to reduce the noise and tolerance of the output voltage. Ideally, we should apply this method over an infinite number of voltage sources, but it also

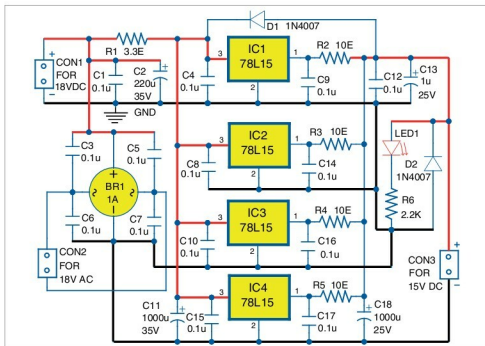


Fig. 1: Circuit diagram of low-noise power supply with four voltage regulators

Different Parameters From The Circuit With Four 78L15 Regulators in Parallel, +18V DC Input Voltage, and R2 Through R5 of 10-ohm Each

Table 1

Parameter	Rload=100-ohm	Rload=80-ohm	Rload=60-ohm
V1	15.02V	15.03V	15.03V
V2	15.08V	15.09V	15.09V
V3	15.06V	15.06V	15.04V
V4	15.03V	15.09V	15.03V
Vout (total)	14.62V	14.55V	14.40V
Iout (total)	146mA	182mA	240mA
Iout (average per regulator)	36.5mA	45mA	60mA
Average voltage over equalisation resistors	0.37V	0.45V	0.60V

works over a small number of voltage regulators. Here we will see a solution based on this method.

Circuit and working

Circuit diagram of the low-noise power supply with four 78L15 voltage regulators in parallel is shown in Fig. 1. It is built around bridge regulator BR1, four 78L15 voltage regulators

(IC1 through IC4), 1N4007 diodes (D1 and D2), LED1, and a few other components.

The circuit uses 18V AC input or 18V DC inputs with RC filters to reduce noise. The RC filters for 18V DC input comprise of resistor R1 and capacitors C1, C2, and C11. The 18V AC input comprises rectifier bridge BR1 and capacitors C1, C2, C3, C5, C6,